# The logarithm function

## Lecture and worked examples

Maths Explorer

# Objectives

Use the properties of ln; solve equations and thresholds; differentiate and study logarithmic functions.

## Definition and properties

The exponential is continuous and strictly increasing on $\mathbb{R}$, and its range is $]0, +\infty[$. For every $x > 0$, there is a unique real number $y$ such that $e^y = x$: it is denoted $\ln x$.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x > 0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \frac{1}{2} \ln a.$$

## Definition and properties

$e^{\ln a+\ln b} = e^{\ln a}e^{\ln b} = ab$; the uniqueness of the preimage gives the formula. There is no analogous formula for $\ln(a + b)$.

## Derivative, variations and limits

We accept that ln is differentiable on $]0, +\infty[$. Differentiating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; hence $(\ln x)' = \dfrac{1}{x} > 0$. Ln is strictly increasing and concave, since its second derivative is $-\dfrac{1}{x^2}$.

$$\lim_{x \to 0^+} \ln x = -\infty, \qquad \lim_{x \to +\infty} \ln x = +\infty.$$

Its curve is the reflection of that of the exponential in the line $y = x$. The line $x = 0$ is a vertical asymptote. Its tangent at 1 gives $\ln x \leq x - 1$ for $x > 0$.

## Derivative, variations and limits

If $u > 0$ is differentiable, $(\ln u)' = \dfrac{u'}{u}$. For every integer $m \geq 1$,

$$\lim_{x \to +\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x \to 0^+} x^m \ln x = 0.$$

## Derivative, variations and limits

Set $t = -\ln x$; as $x \to 0^+$, $t \to +\infty$ and $x \ln x = -te^{-t} \to 0$ by comparative growth. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. At $+\infty$, set $t = \ln x$ to write $\dfrac{\ln x}{x^m} = \dfrac{t}{e^{mt}} \to 0$.

## Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

## Equations and thresholds

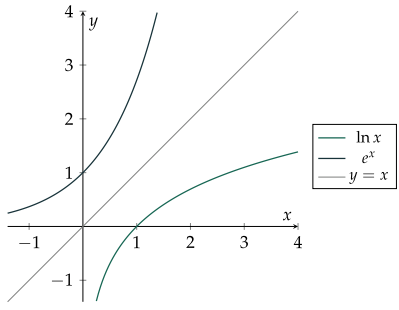
$\ln(x-1)+\ln(x+1)=\ln 8$ requires $x>1$. The equation becomes $x^2-1=8$, so $x=3$ only.

## Equations and thresholds

$0.8^n \leq 0.1$ is equivalent to $n \ln 0.8 \leq \ln 0.1$, then $n \geq \dfrac{\ln 0.1}{\ln 0.8} \simeq 10.319$. The first integer is 11.

# Equations and thresholds

For $f(x) = \dfrac{\ln x}{x}$, $f' = \dfrac{1 - \ln x}{x^2}$; $f$ increases on $]0, e]$, decreases on $[e, +\infty[$, and its maximum is $\dfrac{1}{e}$. Its limit at $0^+$ is $-\infty$, its limit at $+\infty$ is 0.



# Research activity

**Controlling a calculation error - 30 min.**

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Let $e_n$ denote the error after $n$ steps, for $n \in \mathbb{N}$.

## Research activity (2)

1. Express $e_n$ in terms of $n$.
2. Using the logarithm, find the smallest integer $n$ such that $e_n \leq 10^{-4}$.
3. Using a calculator, compute the error at the value found and at the previous one to check that it is the minimum.
4. Now require an error of at most $10^{-6}$. Find the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?